Higher-order galaxy bias for DESI thin-bin angular clustering

A theoretical study for the lssty1xdesi project

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Abstract

The lssty1xdesi forecast combines LSST Y1 3x2pt with DESI angular galaxy clustering C_ℓ^{gg} in thin redshift bins, computed without the Limber approximation. Its likelihood models galaxy clustering with linear bias times the halofit matter power spectrum, and the draft paper extends that to a one-loop Eulerian bias expansion. This note collects the physics that any higher-order bias implementation must respect: the one-loop galaxy power spectrum in real and redshift space, the Eulerian and Lagrangian bias bases with every conversion factor, the DESI DR1 full-shape parameterization and priors, the projection of each term to thin-bin angular spectra, and the measured size of each term at the project's scale cuts. Two conclusions carry the most weight. First, the draft paper's bias table holds the DESI DR1 "physical" sampling combinations $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, inserted into the Eulerian calculator without removing the σ_8 powers, without rotating from the Lagrangian to the Eulerian basis, and without the factor of two of the CCL normalization. The linear-bias column confirms this reading for four of the five chunks; the BGS entry does not match the project fiducial and needs a check against the DESI chains. Under this reading the tidal term enters the galaxy power spectrum with the opposite sign and several times the magnitude of the co-evolution expectation, which moves the loop correction by several percent of P_{gg} at $k \simeq 0.1\,h\,\mathrm{Mpc}^{-1}$, the size of the whole effect. Second, at the project cut $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ the one-loop terms are a 1–3% correction that a minimal one-loop model (the correlated-prior note's per-bin linear bias, co-evolution-centered b_2 and b_{s^2}, one counterterm, one stochastic amplitude per tracer chunk) covers; at the shipped mask near $0.18\,h\,\mathrm{Mpc}^{-1}$ the corrections reach 10–20%, so b_2 and b_{s^2} must be freed per chunk with wide priors and the redshift-space pieces, which this project has not yet modeled, must enter the model.

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1 Scope and context

The lssty1xdesi project forecasts DESI spectroscopic galaxy clustering as angular power spectra C_ℓ^{gg} in 76 thin bins of equal comoving width (about 62 Mpc), combined with LSST Y1 3x2pt. The thin bins resolve radial Fourier modes up to $k_\parallel \simeq \pi/\Delta\chi \simeq 0.05\,\mathrm{Mpc}^{-1} \simeq 0.075\,h\,\mathrm{Mpc}^{-1}$, where $\Delta\chi$ is the comoving bin width, so the angular analysis recovers most of the three-dimensional information of a power spectrum analysis at the same scale cut [11]. The shipped likelihood models the galaxy power as $P_{gg} = b_1^2(z)\,P_{\rm mm}(k,z)$ with $P_{\rm mm}$ the halofit matter power spectrum, one bias amplitude per tracer chunk evolving as $b_1(z) = B_g/D(z)$, where $D(z)$ is the linear growth factor normalized to $D(0) = 1$ and B_g is the chunk amplitude. The $1/D$ shape keeps the clustering amplitude $b_1(z)D(z)$ of a chunk constant; it is the constant-clustering-amplitude shape, not the evolution law of a conserved galaxy population. (The likelihood itself reads one bias parameter per thin bin; the shipped parameter files tie them to one sampled value per chunk.) The project's correlated-prior note (`notes/correlated_bias_prior_v2`, sections "The models" and "Which L and σ_b are physical") shows that this growth-tied model (one amplitude

per group, the bias following $1/D(z)$) is too rigid, and the recommendation of Section 9 uses that note's per-bin model, $b_{1,i} = B_g(1+\delta_i)/D(z_i)$ with one fractional deviation δ_i per bin, split into a free tilt per chunk and a residual under a correlated prior (Section 8).

An earlier review of the forecast (the project reference `references/fom_and_bias_review.md`) concluded that this model is too rigid for the shipped scale cuts: linear bias is accurate only to $k_{\rm max} \simeq 0.08$–$0.12\,h\,{\rm Mpc}^{-1}$, while the shipped masks reach $k \simeq 0.18\,h\,{\rm Mpc}^{-1}$. The missing terms are the second-order and tidal bias loops, the k^2 counterterms, and the stochastic terms. Throughout this note, higher-order bias means the bias beyond linear order: the quadratic b_2 and tidal b_{s^2} at second order and $b_{\rm 3nl}$ at third, all of which enter P_{gg} at one loop. This note is the theoretical study that the higher-order bias implementation will build on. It answers six questions: the structure of the one-loop galaxy power spectrum and its bias bases (Section 2, derivations in Appendix A); the DESI DR1 full-shape model and priors (Section 3); the origin of the draft paper's bias values and the tidal-bias verdict (Section 4); the projection of each term to thin-bin angular spectra (Section 5); the measured size of every term (Section 6); and the recommended model (Sections 8 and 9). Section 10 reports the companion implementation study: the CosmoLike inventory, the comparison with the draft's data vector, and the ordered plan.

Conventions. k is the comoving wavenumber, quoted in h/Mpc unless a unit is written; z is redshift; μ is the cosine between the wavevector and the line of sight; $f = d\ln D/d\ln a$ is the linear growth rate; $\sigma_8(z)$ is the rms linear density fluctuation in spheres of $8\,h^{-1}$Mpc; A_s is the primordial scalar amplitude; $\bar{n}$ is the tracer number density. The matter overdensity is δ, the galaxy overdensity δ_g, and $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\,\delta$ is the tidal field with $s^2 = s_{ij}s_{ij}$.

2 The one-loop galaxy power spectrum

2.1 Real space, Eulerian basis

At one loop, every rotationally invariant local gravitational observable up to third order in the linear density enters the galaxy density. After renormalization, the McDonald–Roy basis [1, 2] carries four deterministic bias parameters plus a derivative bias and a noise field:

$$\delta_g = b_1\,\delta + \frac{b_2}{2}\left(\delta^2 - \langle\delta^2\rangle\right) + \frac{b_{s^2}}{2}\left(s^2 - \langle s^2\rangle\right) + \frac{b_{\rm 3nl}}{2}\,\psi + b_{\nabla^2}\,\nabla^2\delta + \epsilon, \tag{1}$$

where ψ is the one independent third-order operator that survives at one loop (Appendix A writes it out) and ϵ is the stochastic field uncorrelated with the long modes. This is the convention of FAST-PT and of CCL's `EulerianPTCalculator`, verified in this study against the CCL source (`pyccl/nl_pt/ept.py`): the code documents the expansion of Eq. (1) and assembles the auto spectrum as

$$\begin{aligned} P_{gg}(k) = {} & b_1^2\,P_{\rm mm}(k) + b_1 b_2\,P_{\delta_1\delta_2}(k) + \tfrac{1}{4}b_2^2\,P_{\delta_2\delta_2}(k) + b_1 b_{s^2}\,P_{\delta_1 s_2}(k) \\ & + \tfrac{1}{2}b_2 b_{s^2}\,P_{\delta_2 s_2}(k) + \tfrac{1}{4}b_{s^2}^2\,P_{s_2 s_2}(k) + b_1 b_{\rm 3nl}\,\sigma_3^2(k)\,P_{\rm lin}(k) + b_1 b_{k^2} k^2 P(k), \end{aligned} \tag{2}$$

plus stochastic terms discussed below. The six quadratic templates P_{XY} are the mode-coupling integrals of two linear spectra over the second-order kernels (Appendix B); $\sigma_3^2(k)$ is the third-order filter of [1, 2]; and b_{k^2} is CCL's name for the derivative-bias coefficient, whose term is degenerate at this order with the effective-field-theory counterterm $\alpha_0 k^2 P_{\rm lin}$. The draft paper's six-term expression is exactly the first six terms of Eq. (2), as the earlier review already established; the draft then adds $\alpha_0 k^2 P_{\rm lin}$ and sets $b_{\rm 3nl} = 0$.

Three templates do not vanish at low k: $P_{\delta_2\delta_2} \to 2\sigma_4$, $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$ and $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$, with σ_4 a constant built from the linear spectrum (Appendix B). These constants are white noise, indistinguishable from shot noise, and CCL subtracts them only when its `sub_lowk` option is on (the default is off). Whichever setting the pipeline uses must be stated, because with `sub_lowk` off and no free shot-noise parameter the constants enter the data vector as unmodeled power: at the co-evolution biases of the BGS chunk the three constants sum to about 920 $(h^{-1}\mathrm{Mpc})^3$ at $z = 0.3$ (the $\frac{1}{4}b_2^2$ piece alone is 517), five times that chunk's Poisson power $1/\bar{n} \simeq 170$ $(h^{-1}\mathrm{Mpc})^3$. CosmoLike's galaxy-clustering integrand subtracts exactly these constants (Section 10), the `sub_lowk`-on convention, so a CCL vector at the CCL default and a CosmoLike vector differ by this white noise until the two settings are aligned. Section 5.4 follows the subtraction to high k, where it leaves a negative constant.

The hybrid choice of Eq. (2) is that the b_1^2 term carries the full nonlinear matter power (halofit here) while the loop templates stay at one-loop standard perturbation theory. Nicola et al. [9] validated this hybrid against simulations for an LSST-like projected 3x2pt analysis: with the nonlinear $P_{\rm mm}$ in the b_1^2 term the Eulerian model recovers unbiased cosmology to at least $k_{\rm max} = 0.4\,\mathrm{Mpc}^{-1}$, while the fully perturbative $P_{\rm mm}$ fails beyond about $0.2\,\mathrm{Mpc}^{-1}$. The same work finds significant non-Poisson stochasticity in every galaxy sample it fits, a point Section 6 returns to.

2.2 The Lagrangian basis and the exact conversions

The velocileptors code, the DESI DR1 baseline, expands the galaxy field in the initial (Lagrangian) coordinates [5, 6]:

$$F[\delta_0(\mathbf{q})] = 1 + b_1^{\rm L}\,\delta_0 + \frac{b_2^{\rm L}}{2}\left(\delta_0^2 - \langle\delta_0^2\rangle\right) + b_s^{\rm L}\left(s_0^2 - \langle s_0^2\rangle\right) + b_3^{\rm L}\,O_3(\mathbf{q}), \tag{3}$$

where δ_0 and s_0 are the linear density and tidal fields at the initial time and O_3 is the one independent cubic operator. Note the normalization: the quadratic term carries a $1/2$, the tidal term does not. Chen, Vlah and White [5] give the rotation to their Eulerian parameters c_n, defined by $\delta_g = c_1\delta + \frac{c_2}{2}\delta^2 + c_s s^2 + \ldots$ (again no $1/2$ on the tidal term):

$$c_1 = 1 + b_1^{\rm L}, \qquad c_2 = b_2^{\rm L} + \frac{8}{21}\,b_1^{\rm L}, \qquad c_s = b_s^{\rm L} - \frac{2}{7}\,b_1^{\rm L}, \qquad c_3 = b_3^{\rm L} + a\,b_1^{\rm L}, \tag{4}$$

with a a constant fixed by the choice of third-order operator. Appendix A derives the $8/21$ and $-2/7$ from the second-order co-evolution solution. Mapping the c_n onto the CCL convention of Eq. (1) costs one more step, the factor of two between $c_s s^2$ and $(b_{s^2}/2)s^2$:

$$\boxed{b_1 = 1 + b_1^{\rm L}, \qquad b_2 = b_2^{\rm L} + \frac{8}{21}\,b_1^{\rm L}, \qquad b_{s^2} = 2\,b_s^{\rm L} - \frac{4}{7}\,b_1^{\rm L}, \qquad b_{\rm 3nl} = \frac{32}{315}\,b_1^{\rm L} + 2\,b_3^{\rm L}.} \tag{5}$$

The $b_{\rm 3nl}$ entry uses the third-order co-evolution result of Saito et al. [2] for the dynamically generated part; the $2b_3^{\rm L}$ normalization depends on the choice of O_3 and is irrelevant here because every analysis in play sets $b_3^{\rm L} = 0$. A third common basis writes the tidal operator as the Galileon $\mathcal{G}_2 = s^2 - \frac{2}{3}\delta^2$ with coefficient $b_{\mathcal{G}_2}$ (written γ_2 by [10]); then $b_{\mathcal{G}_2} = c_s = b_{s^2}/2$ and the quadratic coefficient shifts by $+\frac{4}{3}b_{\mathcal{G}_2}$ (Appendix A).

2.3 Co-evolution and the $b_n(b_1)$ calibrations

If galaxies form with purely local Lagrangian bias ($b_s^{\rm L} = b_3^{\rm L} = 0$) and are then advected by gravity, the Eulerian tidal and third-order biases are generated by the dynamics alone. Setting $b_s^{\rm L} = 0$ in Eq. (5) gives the co-evolution relations [2]

$$b_{s^2} = -\frac{4}{7}\,(b_1 - 1)\,, \qquad b_{3\rm nl} = \frac{32}{315}\,(b_1 - 1)\,. \tag{6}$$

Both are negative-slope in b_1 for the tidal bias: every tracer with $b_1 > 1$ has $b_{s^2} < 0$ in this convention. Simulations support the relations at the 10–30% level; the excursion-set calibration used by [10], $b_{\mathcal{G}_2} = 0.524 - 0.547\,b_1 + 0.046\,b_1^2$, gives a $b_{s^2} = 2b_{\mathcal{G}_2}$ about 30% more negative than Eq. (6) at $b_1 \simeq 2.2$. For the quadratic bias the separate-universe fit of Lazeyras et al. [3], valid for $1 \lesssim b_1 \lesssim 10$ and redshift independent, reads

$$b_2(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3. \tag{7}$$

Figure 1 draws these expectations across the six DESI chunks of this project, with $b_1 = B_g/D(z)$ and the project amplitudes $B_g = 1.34$ (BGS), 1.70 (LRG) and 0.84 (ELG). Each curve spans its chunk, so its slope shows how much the co-evolution b_{s^2} moves across a chunk when b_1 follows the $B_g/D(z)$ shape; this drift is the subject of Section 8.

2.4 Redshift space

In redshift space the one-loop model gains no new bias parameters but several new terms. The DESI DR1 methodology paper [7] writes every full-shape model in the generic form

$$P_{s,g}(k,\mu) = P_{s,g}^{\rm PT}(k,\mu) + \left(\alpha_0 + \alpha_2\mu^2 + \alpha_4\mu^4\right) k^2 P_{\rm lin}(k,\mu) + \left({\rm SN}_0 + {\rm SN}_2\,k^2\mu^2 + {\rm SN}_4\,k^4\mu^4\right). \tag{8}$$

$P_{s,g}^{\rm PT}$ is the one-loop redshift-space spectrum of biased tracers. Its tree-level limit is the Kaiser spectrum $(b_1 + f\mu^2)^2\,P_{\rm lin}$; at one loop it couples every bias operator of Eq. (3) to the velocity field, producing μ-dependent loop terms with the same four bias coefficients (the complete velocileptors expression is the resummed one-loop redshift-space power spectrum in the theory section of [6], built from the displacement-field correlators of [5]). The three counterterms $\alpha_0, \alpha_2, \alpha_4$ absorb the small-scale sensitivity of the loop integrals and the leading velocity dispersion; velocileptors normalizes them as fractional corrections to the linear multipoles, so that $\alpha_n = 12.5\,h^{-2}{\rm Mpc}^2$ is a 50% correction to the n-th moment at $k = 0.2\,h\,{\rm Mpc}^{-1}$ [6]. The three stochastic terms descend from the noise field: ${\rm SN}_0$ is the deviation of the constant noise from the Poisson value already subtracted from the estimator, ${\rm SN}_2$ is the velocity-weighted noise (the leading Fingers-of-God stochastic term), and ${\rm SN}_4$ its next moment.

The structural summary: a one-loop redshift-space analysis carries per tracer sample four deterministic biases ($b_1^{\rm L}, b_2^{\rm L}, b_s^{\rm L}, b_3^{\rm L}$), three counterterms, and three stochastic amplitudes; a real-space or projected analysis keeps the same biases but collapses the counterterms toward α_0 and the stochastic tower toward ${\rm SN}_0$, with the μ-dependent members reappearing in exactly the way Section 5 describes.

3 The DESI DR1 full-shape model and priors

The DESI DR1 full-shape analysis [7, 8] fits the power spectrum monopole and quadrupole of six tracer bins (BGS at $z_{\rm eff} = 0.295$, LRG1–3 at 0.510, 0.706, 0.919, ELG2 at 1.317, QSO at 1.491)

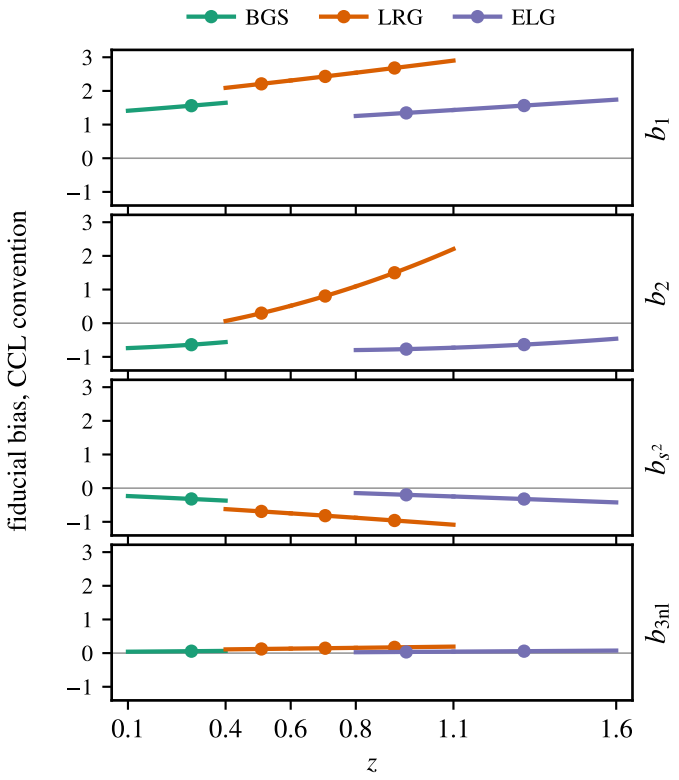


Figure 1: The fiducial biases of the six DESI chunks in the CCL convention of Eq. (1), against redshift: the project's linear bias $b_1 = B_g/D(z)$, the Lazeyras et al. b_2 of Eq. (7), and the co-evolution b_{s^2} and $b_{3\mathrm{nl}}$ of Eq. (6). Each curve spans one chunk (the redshift ticks are the chunk edges); the chunks of one tracer share B_g, so their curves join. Markers sit at the DESI DR1 effective redshifts; ELG1, which the DR1 full-shape analysis excluded, is marked near its chunk's center. D is computed at the project fiducial cosmology (Appendix C).

over $0.02 < k < 0.2\,h\,\mathrm{Mpc}^{-1}$. The ELG1 bin ($0.8 < z < 1.1$) failed the fibre-collision systematics tests and is excluded from the full-shape analysis [7]. Four theory codes were validated against each other and simulations; the baseline is the Eulerian mode of velocileptors, with Folps and PyBird as cross-checks and a Gaussian-streaming configuration-space model as a fourth [7]. The parameter choices that matter here:

- **Basis and sampling.** Even in the Eulerian mode, DESI samples the Lagrangian parameters of Eq. (3), reparameterized as the "physical" combinations

$$\left(1 + b_1^{\mathrm{L}}\right)\sigma_8(z_{\mathrm{eff}}), \qquad b_2^{\mathrm{L}}\,\sigma_8^2(z_{\mathrm{eff}}), \qquad b_s^{\mathrm{L}}\,\sigma_8^2(z_{\mathrm{eff}}), \tag{9}$$

  because the data constrain these products directly and the reparameterization reduces projection effects on σ_8 [7, 8]. Note that $1 + b_1^{\mathrm{L}}$ is the Eulerian linear bias, so the first combination is $b_1\sigma_8(z_{\mathrm{eff}})$.

- **Priors.** Uniform $[0, 3]$ on $(1+b_1^{\mathrm{L}})\sigma_8$; Gaussian with mean zero and width 5 on $b_2^{\mathrm{L}}\sigma_8^2$ and on $b_s^{\mathrm{L}}\sigma_8^2$; Gaussian with mean zero and width $12.5\,h^{-2}\mathrm{Mpc}^2$ on α_0 and α_2 (the 50%-correction-at-k_{max} rule); Gaussian widths of 2 Poisson units on SN_0 and of 5 units of $f_{\mathrm{sat}}\sigma_v^2/\bar{n}$ on SN_2, where f_{sat} is the satellite fraction and σ_v a characteristic velocity dispersion [7, 8].

- **Fixed parameters.** $b_3^{\mathrm{L}} = 0$, because the cubic bias is expected to be small for intermediate-mass halos and is degenerate with the counterterms [7, 6]; α_4 and SN_4 drop with the hexadecapole, which DESI DR1 does not fit. The tidal bias is deliberately left free rather than tied to co-evolution: tying it would add cosmological information that DESI chose not to claim [7].

- **Budget.** Seven nuisance parameters per bin $(b_1^{\mathrm{L}}, b_2^{\mathrm{L}}, b_s^{\mathrm{L}}, \alpha_0, \alpha_2, \mathrm{SN}_0, \mathrm{SN}_2)$, 42 in total, with the counterterms and stochastic terms marginalized analytically [8].

Semenaite et al. [10] fit the same DESI DR1 tracers in configuration space jointly with weak lensing, with one structural difference relevant to this project: both tidal biases are fixed to calibrated relations in b_1 (the excursion-set $\gamma_2(b_1)$ and the conserved-evolution $\gamma_{21}(b_1, \gamma_2)$), leaving b_1 and b_2 free per bin with flat priors, plus counterterms and a virial damping parameter. Their validation on mocks shows that fixing the tidal sector to these relations recovers unbiased cosmology for DESI-like tracers, which is direct evidence for the co-evolution-anchored minimal model this note recommends.

4 The draft paper's bias table: the tidal-bias verdict

The draft paper's bias table (its Table of "bias parameters per DESI redshift bin") lists, per chunk, values labeled b_1^{E}, b_2^{E}, b_s^{E}, attributed to the DESI DR1 full-shape analysis. The earlier review flagged the b_s^{E} column (positive, 0.6–1.4) as incompatible with co-evolution, which predicts b_{s^2} between -0.3 and -1.0 for these tracers (Figure 1), and suspected a Lagrangian-to-Eulerian conversion error without establishing it. The provenance can now be established from the numbers themselves.

Three facts identify the table as the DESI physical-basis combinations of Eq. (9) inserted raw:

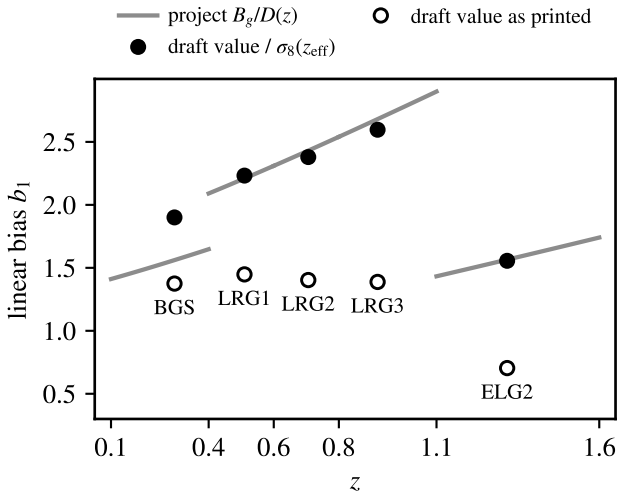


Figure 2: The draft table's linear-bias column ("$b_1^{\rm E}$") read as the DESI physical combination $b_1\sigma_8(z_{\rm eff})$: as printed (open circles) and divided by σ_8 at the chunk's DESI DR1 effective redshift (filled circles), against the project's fiducial $b_1 = B_g/D(z)$ over each chunk (gray). $\sigma_8(z)$ is the rms linear density fluctuation in spheres of $8\,h^{-1}$Mpc at the project fiducial cosmology, $\sigma_8(0) = 0.843$. The draft has no ELG1 row.

1. **The linear column divided by $\sigma_8(z_{\rm eff})$ lands on the known biases for four of the five chunks.** Figure 2 shows the check: for LRG1–3 and ELG2 the implied Eulerian bias agrees with the project's fiducial $B_g/D(z_{\rm eff})$ to 1–5%. An independent check at a Planck-like cosmology ($\Omega_m = 0.3138$, $\sigma_8 = 0.8076$, run by the implementation study) finds the same pattern: ratios of the implied bias to $B_g/D(z_{\rm eff})$ of 1.055, 1.022, 1.011 and 1.038 for LRG1–3 and ELG2. Read directly as Eulerian biases instead, the column is unphysical: it would make the three LRG chunks nearly unbiased ($b_1 \simeq 1.4$ where LRGs sit near 2.2–2.7) and the ELG2 chunk anti-biased ($b_1 = 0.70$) at $z = 1.3$. The BGS row is the exception: it implies $b_1 \simeq 1.90$ against the project's 1.56, a ratio of 1.22, and 1.27 in the Planck-like check. Whether that reflects the project's BGS fiducial $B_g = 1.34$, which comes from the source forecast's configuration rather than from a DESI DR1 fit, or a different origin of that row, is not established here. The direct test is to read the BGS entry of the public DESI DR1 full-shape chains: if the maximum-posterior $(1 + b_1^{\rm L})\sigma_8$ is 1.375, the reading covers BGS as well and the project's BGS amplitude is the quantity to revisit.

2. **The empty ELG1 row.** DESI DR1 excluded ELG1 from the full-shape analysis (Section 3); a table transcribed from the DESI full-shape products has nothing to put there, and the draft's row is indeed empty.

3. **The magnitudes of the other two columns.** Read as $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, the draft's second and third columns (-1.5 to $+0.3$ and $+0.6$ to $+1.4$) sit comfortably inside the DESI Gaussian priors of width 5 on exactly those combinations. Read as CCL Eulerian parameters they contradict both co-evolution and the DESI prior centering.

Verdict. The draft's bias table holds the DESI DR1 sampling combinations $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, used as CCL's Eulerian (b_1, b_2, b_{s^2}) with three transformations still to apply:

removing the σ_8 powers, rotating the Lagrangian parameters to the Eulerian basis via Eq. (5), and the factor of two between the velocileptors and CCL tidal normalizations. The reading is established by the linear column for LRG1–3 and ELG2 and extends to the other two columns by the table's construction; the BGS linear entry awaits the chain check above. The sign puzzle dissolves: the column was never an Eulerian tidal bias. The corrected recipe, if DESI values are wanted, is

$$b_1 = \frac{[(1+b_1^{\rm L})\sigma_8]}{\sigma_8(z_{\rm eff})}, \quad b_2 = \frac{[b_2^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} + \frac{8}{21}(b_1-1), \quad b_{s^2} = \frac{2\,[b_s^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} - \frac{4}{7}(b_1-1), \tag{10}$$

with the bracketed quantities the DESI chain values and σ_8 evaluated at the DESI fiducial cosmology. This note recommends against importing the DESI nuisance values at all, beyond the linear amplitude: applied to the draft's numbers, Eq. (10) gives $b_s^{\rm L} \simeq 2.5$–3.3 and $b_{s^2} \simeq +4$ to $+6$, which says that these particular chain values are dominated by the width of the DESI prior and by degeneracies with the counterterms, not by a measurement of the tidal field. A forecast fiducial should come from the calibrated relations of Section 2.3 (Figure 1); the DESI posteriors then motivate the prior widths, not the central values.

5 Projection to thin-bin angular spectra

5.1 Separability and the D^4 rule

The exact (non-Limber) angular power spectrum between bins i and j is

$$C_\ell^{ij} = \frac{2}{\pi}\int dk\, k^2 \int d\chi_1\, W_i(\chi_1)\, j_\ell(k\chi_1) \int d\chi_2\, W_j(\chi_2)\, j_\ell(k\chi_2)\, \mathcal{P}(k; z_1, z_2), \tag{11}$$

where $W_i(\chi)$ is the bin's radial kernel (the normalized, Fingers-of-God-convolved $n(z)$ expressed in comoving distance χ), j_ℓ is the spherical Bessel function, and $\mathcal{P}$ is the unequal-time galaxy power spectrum. The computation is tractable because every term of the one-loop expansion is separable in k and time:

- the linear term carries $D(z_1)\,D(z_2)$;
- all five quadratic templates ($P_{\delta_1\delta_2}$, $P_{\delta_2\delta_2}$, $P_{\delta_1 s_2}$, $P_{\delta_2 s_2}$, $P_{s_2 s_2}$) are 22-type mode-coupling integrals over two linear spectra, so each carries $D^2(z_1)\,D^2(z_2)$;
- the 13-type pieces (the $\sigma_3^2 P_{\rm lin}$ term of $b_{3\rm nl}$ and the P_{13} inside the one-loop matter spectrum) carry the symmetrized $\frac{1}{2}\left[D^3(z_1)D(z_2) + D(z_1)D^3(z_2)\right]$.

At equal times every loop term scales as $D^4(z)$. This is exactly how both CCL and velocileptors evaluate the model: the loop templates are computed once from the $z = 0$ linear spectrum and rescaled by $D^4(z)$, with the bias coefficients supplying the remaining redshift dependence [8]. For the thin bins of this project the equal-time evaluation at the bin (or pair) mean redshift is accurate: across a $\Delta z = 0.02$ bin the growth factor changes by about 1%, the symmetric products above differ from $D^4(z_{\rm mid})$ only at second order in that 1%, and the residual is a few parts in 10^4 of the loop terms, which are themselves percent-level corrections.

The consequence for the FFTLog machinery is structural, and it is the one fact the implementation study needs from this section: adding the one-loop terms adds a fixed, small set of z-independent k-shapes (the six quadratic templates, $\sigma_3^2 P_{\rm lin}$, $k^2 P_{\rm lin}$, and a constant), each

entering a bin's power spectrum with a scalar amplitude $b_X b_Y D^4(z_{\rm bin})$. The non-Limber cost scales with the number of distinct k-shapes, not with the number of bins. The project's low-rank engine projects the full per-bin $P_{gg}(k)$ through an SVD of its square root, so it carries the one-loop sum with no structural change; only the sign of P_{gg} matters there.

CosmoLike subtracts the low-k constants of the quadratic templates (the σ_4 terms of Section 2.1), so the loop sum vanishes as $k \to 0$ and large scales keep the linear-bias spectrum. As $k \to \infty$ the templates decay, and the same subtraction leaves a constant,

$$P_\infty = -\frac{\sigma_4}{2}\, D^4(z) \left(b_2 + \frac{2}{3}\, b_{s^2}\right)^2 \leq 0, \tag{12}$$

with $\sigma_4 = 4391\,(h^{-1}\mathrm{Mpc})^3$ at $z = 0$ at the project fiducial. Where this negative constant outweighs the decaying positive terms, P_{gg} turns negative at high k, and its square root does not exist there. Section 5.4 derives P_∞ and measures where the sign changes. The low-rank engine first guarded the square root with a floor on P_{gg}; it now handles the constant exactly, by factorizing $P_{gg} - P_\infty$ and projecting P_∞ separately (Section 5.4).

5.2 Linear redshift-space distortions and counterterms for thin shells

Linear (Kaiser) redshift-space distortions enter Eq. (11) through the transformation of the density kernel: each factor of $f\mu^2$ in the flat-sky spectrum becomes a second derivative of the Bessel kernel in the exact projection,

$$\Delta^i_\ell(k) = \int d\chi\, W_i(\chi) \left[b_1(z)\, j_\ell(k\chi) - f(z)\, j''_\ell(k\chi)\right] D(z), \tag{13}$$

the form the project's covariance already implements. For wide photometric bins the j''_ℓ term integrates down by orders of magnitude; for shells thinner than the scales probed it survives at full strength, because the shell resolves line-of-sight modes with μ near one. This is why thin-bin C^{gg}_ℓ recovers the Kaiser information of a three-dimensional analysis [11], and it cuts both ways: every μ-dependent term of Eq. (8) that a wide-bin analysis could drop is alive here.

The counterterms follow the same rule. The $\alpha_0 k^2 P_{\rm lin}$ piece is μ-independent and projects like the density: one extra k-shape, $k^2 P_{\rm lin}(k)$, with per-bin amplitude $\alpha_0 D^2(z_{\rm bin})$ (the counterterm multiplies the linear spectrum, so it scales as D^2, not D^4). The $\alpha_2 \mu^2 k^2 P_{\rm lin}$ piece projects through one j''_ℓ in one of the two Bessel kernels, the same transfer structure as the Kaiser term; for thin shells its reach is comparable to α_0's rather than suppressed. The project's Fingers-of-God convolution of each bin's $n(z)$ with the velocity-dispersion kernel σ_v is a nonperturbative stand-in for part of this physics: it damps radial power with a μ-dependent, scale-dependent shape set by the sampled σ_v, and is therefore partially degenerate with α_2. Partial degeneracy is not identity: the convolution is a positive damping of fixed functional form, while α_2 can take either sign and scales as $k^2 P_{\rm lin}$.

5.3 Stochastic terms under projection

A constant three-dimensional power SN_0 projects to

$$C^{ij}_\ell\big|_{\rm stoch} = \mathrm{SN}_0 \int d\chi\, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2}, \tag{14}$$

independent of ℓ. Equation (14) is exact, not a Limber approximation: for a constant $\mathcal{P}$ the k integral of Eq. (11) collapses by Bessel orthogonality, so the Limber formula is exact for white noise. For a top-hat bin of width $\Delta\chi$ the result is $\mathrm{SN}_0/(\chi^2\Delta\chi)$, and at the Poisson value $\mathrm{SN}_0 = 1/\bar{n}$ it reproduces the familiar angular shot noise $1/\bar{n}_{2D}$, with $\bar{n}_{2D}$ the surface density of the bin. The angular pipeline already carries the Poisson part in the covariance as the noise spectrum N_ℓ, so the free parameter in the mean model is the deviation from Poisson, not the full $1/\bar{n}$. Two projection-specific effects deserve attention. First, the Fingers-of-God convolution makes adjacent thin-bin kernels overlap, so Eq. (14) is nonzero for neighboring pairs: a stochastic term leaks into the cross spectra that the project measures, with a magnitude set by the kernel overlap integral. Second, the $\mathrm{SN}_2 k^2\mu^2$ term projects with the radial-derivative structure of Section 5.2 and grows as k^2; Section 6 shows it stays below the other terms at this project's cuts.

5.4 The white-noise limit of the renormalized loops

Origin of the constant. The template $P_{\delta_2\delta_2}$ is the power spectrum of the operator δ^2. Its four linear fields pair across the two operators in two ways (two Wick contractions), and each pairing gives $\int_{\mathbf{q}} P_{\rm lin}(q)\, P_{\rm lin}(|\mathbf{k}-\mathbf{q}|)$. As $k \to 0$ this integral tends to $\sigma_4 = \int_{\mathbf{q}} P_{\rm lin}^2(q)$, so $P_{\delta_2\delta_2} \to 2\sigma_4$. The constant comes from pairs of nearly opposite wavevectors, $\mathbf{q}$ and $\mathbf{k}-\mathbf{q} \simeq -\mathbf{q}$, whose sum is the long mode; on scales much larger than $1/q$ it is indistinguishable from white noise. The tidal templates carry the kernel $S_2 = \mu_{12}^2 - \frac{1}{3}$ of Appendix B, which equals $\frac{2}{3}$ for opposite wavevectors ($\mu_{12} = -1$): one power of it gives $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$, two give $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$.

Subtracting $2\sigma_4$, $\frac{4}{3}\sigma_4$ and $\frac{8}{9}\sigma_4$ from the $\delta^2\delta^2$, $\delta^2 s^2$ and $s^2 s^2$ templates renormalizes the operators: the constant moves into the stochastic sector, and the loops vanish on large scales. As $k \to \infty$ the templates themselves decay, so the subtracted loop sum of Eq. (2) tends to minus the subtracted constants, each with its bias coefficient:

$$\tfrac{1}{4}b_2^2\,(2\sigma_4) + \tfrac{1}{2}b_2 b_{s^2}\,(\tfrac{4}{3}\sigma_4) + \tfrac{1}{4}b_{s^2}^2\,(\tfrac{8}{9}\sigma_4) = \frac{\sigma_4}{2}\left(b_2^2 + \tfrac{4}{3}b_2 b_{s^2} + \tfrac{4}{9}b_{s^2}^2\right) = \frac{\sigma_4}{2}\left(b_2 + \tfrac{2}{3}b_{s^2}\right)^2 . \qquad (15)$$

With the D^4 scaling of Section 5.1, this square with a minus sign is P_∞ of Eq. (12). The perfect square has a reading: $b_2 + \frac{2}{3}b_{s^2}$ is the quadratic coefficient in the Galileon basis of Appendix A, and the Galileon kernel $S_2 - \frac{2}{3} = \mu_{12}^2 - 1$ vanishes for opposite wavevectors, so $\mathcal{G}_2$ adds no constant of its own.

Two conventions. CosmoLike subtracts the three constants, as CCL does with `sub_lowk` on. The velocileptors code, the DESI DR1 baseline in both its Eulerian (EPT) and Lagrangian (LPT) flavors, keeps the loop proportional to the square of the quadratic bias unsubtracted. In LPT the term is $\frac{1}{2}(b_2^{\rm L})^2\xi_{\rm lin}^2$ in configuration space, with $\xi_{\rm lin}$ the linear correlation function; in EPT it is $(c_2^2/2)\int_{\mathbf{q}} P_{\rm lin}P_{\rm lin}$, with c_2 the Eulerian quadratic bias of Eq. (4). The code removes only the operator means $\langle\delta^2\rangle$ and $\langle s^2\rangle$ of Eq. (3), which touch only the $k = 0$ mode, and the stochastic amplitude SN_0 of Eq. (8) absorbs the low-k constant [5, 6].

The two conventions are the same model. The subtracted spectrum with SN_0 centered at zero equals the unsubtracted spectrum with SN_0 centered at $-(\sigma_4/2)D^4(b_2 + \frac{2}{3}b_{s^2})^2$, a center that moves with the biases. A prior on SN_0 with a fixed center therefore states different physics in the two conventions.

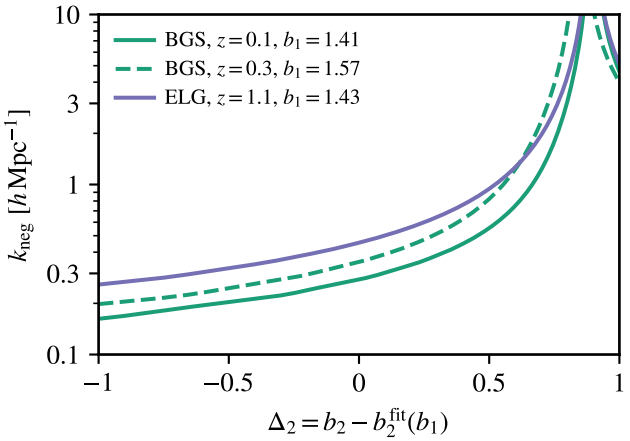


Figure 3: The wavenumber $k_{\rm neg}$ at which the one-loop P_{gg} in CosmoLike's subtracted convention first turns negative, against the offset Δ_2 of b_2 from the Lazeyras calibration, for two BGS bins and one ELG bin. The tidal bias b_{s^2} follows co-evolution, $b_1 = B_g/D(z)$ with the project amplitudes, and $b_{3\rm nl}$ and the counterterm are zero. The leading term is $b_1^2 D^2 P_{\rm lin}$; the halofit $P_{\rm mm}$ that the likelihood puts in that term is larger at these wavenumbers, so the hybrid model of Eq. (2) turns negative at somewhat higher k. Where $b_2 + \frac{2}{3} b_{s^2}$ vanishes, P_∞ of Eq. (12) vanishes with it and $k_{\rm neg}$ leaves the panel. The numerical set-up is in Appendix C.

Where P_{gg} turns negative. Let $k_{\rm neg}$ be the wavenumber at which the subtracted one-loop P_{gg} first turns negative, and let $\Delta_2 = b_2 - b_2^{\rm fit}(b_1)$ be the offset of b_2 from the Lazeyras calibration of Eq. (7) (the group offset $\Delta_{2,g}$ of Section 8.1). Figure 3 draws $k_{\rm neg}$ against Δ_2 for three representative bins, with b_{s^2} on co-evolution.

Three facts frame the figure. In the LRG bins the constant stays small against the galaxy power, because the calibrated b_2 partly cancels $\frac{2}{3} b_{s^2}$ and the larger b_1^2 raises the positive terms: $k_{\rm neg} > 1.5\,h\,{\rm Mpc}^{-1}$. At the calibrated biases ($\Delta_2 = 0$), $|P_\infty|$ is 5–42 times the Poisson power $1/\bar{n}$ in the forecast's BGS bins, where $1/\bar{n} \simeq 33$–$170\,(h^{-1}{\rm Mpc})^3$, and 3–15% of it in the ELG bins. Only negative offsets matter: in these bins $b_2 + \frac{2}{3} b_{s^2}$ vanishes near $\Delta_2 \simeq +0.9$, so a positive offset shrinks the constant and pushes $k_{\rm neg}$ far out.

Why thin-bin angular spectra see it and DESI's 3D fits do not. An angular spectrum integrates $P(k)$ over all k through each bin's radial window (Eq. (11)). A thin bin passes radial modes far above the transverse wavenumber of a band, so no cut in ℓ excludes the region $k > k_{\rm neg}$; the UV-fraction criterion only bounds its share. A white noise projects exactly, to $P \int d\chi\, W_i W_j/\chi^2$ (Eq. (14)), so in the subtracted convention P_∞ enters every multipole at full strength.

DESI's full-shape fits are spared both effects. They use only $0.02 < k < 0.20\,h\,{\rm Mpc}^{-1}$ in three dimensions [7], so the region beyond $k_{\rm neg}$ does not enter them. And the DESI samples are sparse. DR1 BGS holds 300,017 galaxies over $0.1 < z < 0.4$; that count and the survey area give $1/\bar{n} \approx 3 \times 10^3\,(h^{-1}{\rm Mpc})^3$, an estimate made for this note rather than a published DESI value. The constant is then at most of the order of the Poisson power and fits DESI's ${\rm SN}_0$ prior, $\mathcal{N}(0, 2^2)/\bar{n}$, in either convention. The forecast's BGS is 20–100 times denser.

What CosmoLike does. The low-rank engine splices its non-Limber density term into the Limber spectrum: it takes the full Limber C_ℓ, subtracts the Limber projection of the density-only term, and adds the non-Limber projection of the same term. Since the cosmolike_core change that splits the white-noise constant off the factorized spectrum, the constant passes through this splice exactly:

- the engine factorizes $P_{gg} - P_\infty$, the spectrum with the unsubtracted templates, which is positive, so its square root exists;
- the Limber density-only term subtracted in the splice uses the same unsubtracted templates;
- the full Limber term keeps the subtracted templates, so P_∞ reaches C_ℓ through Limber, which is exact for a constant power (Section 5.3);
- a cross pair of bins i and j carries the bilinear constant $-(\sigma_4/2)D^4(b_{2,i}+\frac{2}{3}b_{s^2,i})(b_{2,j}+\frac{2}{3}b_{s^2,j})$.

Before this change the engine clamped P_{gg} to a small positive floor on about a third of its (k, z) grid. The floor stays in the code as a guard, and the tested vectors no longer reach it. The measured effects, on the project's data vectors:

- linear-bias vectors, and a one-loop vector computed with the FKEM engine, are bitwise unchanged;
- at the calibrated biases the low-rank one-loop vector moves by at most 5×10^{-3} of the angular shot noise;
- at the highest band the model A cut keeps, the share of a bin's white-noise projection that comes from $k > k_{\rm neg}$ is at most 0.34% (BGS, $z = 0.11$) and falls to about zero for ELG (a flat-sky estimate at the calibrated biases);
- one model A prior width below the calibrations ($\Delta_2 = -1$ and $\Delta_s = -0.5$, with Δ_s the offset of b_{s^2} from co-evolution), the clamp had hidden negative C_ℓ in the lowest BGS bins beyond $k \approx 0.15\,h\,{\rm Mpc}^{-1}$, in the Limber correspondence $k = (\ell + \frac{1}{2})/\chi$. An example is -29 times the shot noise at $\ell = 50$ on the shipped mask; the FKEM engine, untouched by the change and Limber in its one-loop content, gives -25;
- at the model A cut the same biases keep every kept C_ℓ positive, and the exact treatment moves them by at most 0.27 of the shot noise.

Consequences and remedies. Model A, at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$, is safe: at the calibrated biases the region beyond $k_{\rm neg}$ carries a negligible share of its kept bands, and one prior width below the calibrations every kept C_ℓ stays positive. Model B, at $k_{\rm max} \approx 0.18\,h\,{\rm Mpc}^{-1}$ on the dense BGS, is not. For the low-redshift BGS bins, one-loop perturbation theory is not trustworthy near $k \approx 0.15$–$0.2\,h\,{\rm Mpc}^{-1}$ once b_2 sits a unit below the calibration (Figure 3). Three remedies are available:

1. A BGS-specific cut, $k_{\rm max} \approx 0.14\,h\,{\rm Mpc}^{-1}$ at $z = 0$. It contains the problem but does not remove it.

2. A taper on the loop terms above the scale where one-loop perturbation theory is trusted, so that $P_{gg} \to b_1^2 P_{\rm mm}$ at high k for every b_2. It removes the problem at the cost of changing the model in the ultraviolet, a change the UV-fraction cut bounds. For low-redshift BGS the taper would sit near $0.15\,h\,\mathrm{Mpc}^{-1}$.

3. For chains, a positivity prior that rejects points with $P_{gg} < 0$ below a chosen k, like the project's amplitude cut (the prior that keeps $(A_s/A_{s,\rm fid})(b_i/b_{i,\rm fid})^2 \leq 8$ per group, with b_i a bin's linear bias). It leaves the model unchanged and does not affect a Fisher forecast.

Two other levers miss the sign problem. A tighter b_2 prior only shifts $k_{\rm neg}$. An SN_0 prior in absolute units, $(h^{-1}\mathrm{Mpc})^3$ instead of multiples of $1/\bar{n}$, does not touch the negativity, because SN_0 adds the same constant at every k and leaves the one-loop P_{gg} as it is. It solves a separate problem: in the unsubtracted convention SN_0 must absorb a constant of 5–42 Poisson units in the forecast's BGS, and a width of $2/\bar{n}$ is too narrow for a sample this dense.

5.5 What an angular analysis can drop

Terms that can be dropped at both of this project's cuts: the μ^4 members (α_4, SN_4), which DESI itself drops with the hexadecapole and whose thin-shell projections are suppressed relative to their μ^2 partners; the $b_{3\rm nl}$ term, bounded by co-evolution at $|b_{3\rm nl}| \leq 0.17$ and worth at most 0.6% of P_{gg} at $k \leq 0.1\,h\,\mathrm{Mpc}^{-1}$ (Figure 4), and degenerate with α_0 in shape over the fitted range; and the derivative bias b_{k^2}, exactly degenerate with α_0 at one loop. Terms that cannot be dropped at the shipped mask: the b_2 and b_{s^2} loops, α_0, a radial counterterm degree of freedom (an α_2 equivalent or a demonstrably sufficient σ_v freedom), and SN_0. The quantitative basis follows.

6 The size of every term

Figure 4 draws the fractional contribution of each one-loop term to P_{gg} against k, computed with FAST-PT 4.0 and CAMB 1.6.5 in the group's `ccl` conda environment at the project fiducial cosmology (the `cocoapy311` environment carries neither package; Appendix C records the setup). The reference is the linear-bias model $b_1^2 P_{\rm mm}$ with halofit, the quantity the current likelihood computes. Biases follow Figure 1 at each tracer's redshift; the dashed curve shows the counterterm at the one-sigma DESI prior $\alpha_0 = 12.5\,h^{-2}\mathrm{Mpc}^2$ as a size scale. Figure 5 shows, on the same scale, the Poisson power $1/\bar{n}$ from the project's own thin-bin densities, which the covariance carries as noise, as context for the stochastic freedom.

Reading the figures at the project's two reference cuts:

At $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ (the non-Limber UV-fraction cut that matches the thin-bin radial resolution), the net loop correction is 1–3% of P_{gg}, with individual terms up to 3% and partial cancellations that depend on the bias values. The counterterm prior scale is 1–3%. A 30% non-Poisson stochastic residual, the size Nicola et al. [9] detect for red samples, is 3% of P_{gg} for the LRG chunks and more for ELG. Against the forecast's statistical power (sub-percent amplitude precision per chunk once all bins and bands are aggregated), none of these is negligible; the saving grace is that below $0.075\,h\,\mathrm{Mpc}^{-1}$ their scale dependence is mild, so the free linear amplitudes absorb most of each term and the damage concentrates in the residual shape. That is an argument for a small extension, not for linear bias alone.

At the shipped mask, $k \simeq 0.18\,h\,\mathrm{Mpc}^{-1}$, the picture changes qualitatively. The $b_1 b_2$ term alone reaches 7–17% with strong scale dependence; the quadratic-quadratic terms reach 3–7%;

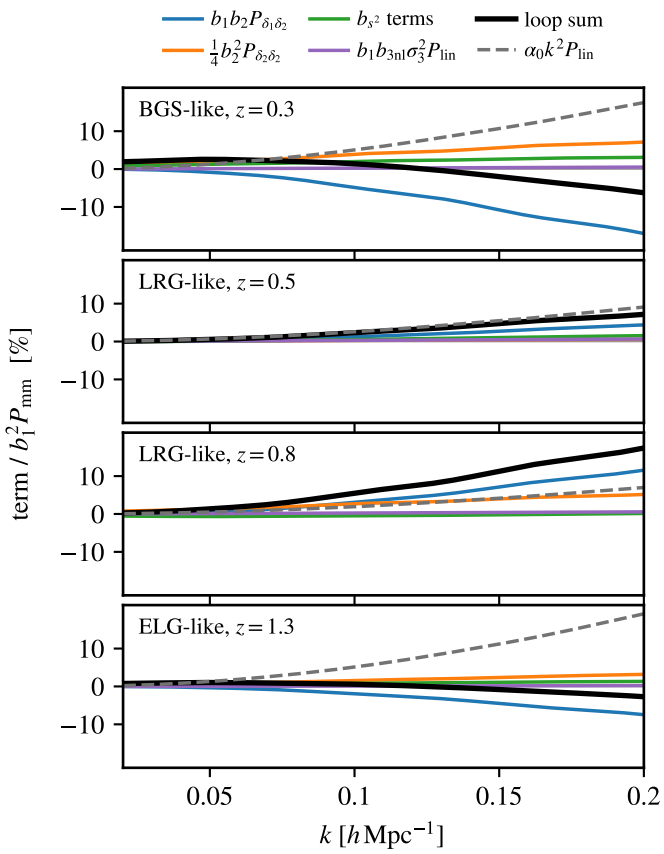


Figure 4: Each one-loop term of Eq. (2) in percent of the linear-bias model $b_1^2 P_{\mathrm{mm}}$ (halofit), against k at the tracer redshift, for four DESI-like tracers at the co-evolution biases of Figure 1. The three tidal terms ($b_1 b_{s^2} P_{\delta_1 s_2}$, $\frac{1}{2} b_2 b_{s^2} P_{\delta_2 s_2}$ and $\frac{1}{4} b_{s^2}^2 P_{s_2 s_2}$) are drawn as their sum. Loop templates are FAST-PT at $z = 0$ scaled by $D^4(z)$, with the low-k constants kept (CCL's default, `sub_lowk` off). The dashed curve is the counterterm $\alpha_0 k^2 P_{\mathrm{lin}}(k, z)$ at the DESI one-sigma prior width $\alpha_0 = 12.5\, h^{-2}\mathrm{Mpc}^2$.

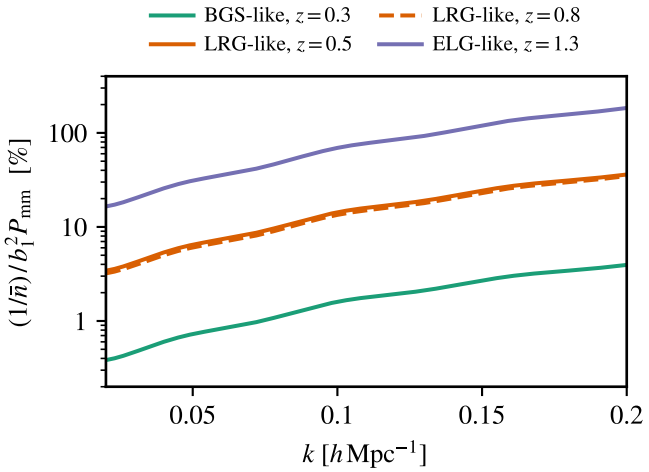


Figure 5: The Poisson power $1/\bar{n}$ in percent of the linear-bias model $b_1^2 P_{\rm mm}$ (halofit) of the four tracers of Figure 4, with $\bar{n}$ the density of the project's DESI DR3-forecast thin bin nearest each tracer redshift. The covariance carries this power as noise; it is not part of the model and is drawn to set the scale of the stochastic freedom.

the counterterm scale is 10–19%. These are not absorbable by an amplitude, and their shapes differ from one another, so the higher-order biases must be free per chunk rather than tied to the calibrations, as the DESI DR1 choice of $k_{\rm max} = 0.2\,h\,{\rm Mpc}^{-1}$ with seven nuisances per tracer already reflects.

7 The draft model against DESI's

Point by point, the draft's Eulerian model against the DESI DR1 full-shape baseline it cites:

- **The six quadratic templates match CCL exactly**, and CCL matches the McDonald–Roy structure that FAST-PT implements. No issue beyond the `sub_lowk` statement of Section 2.1.
- **The bias values need the conversion of Section 4** before they can enter an Eulerian calculator, and this is the largest single correction. At the table's numbers the tidal term enters with the opposite sign and several times the magnitude of the co-evolution expectation (the BGS $b_1 b_{s^2}$ product is $+1.96$ against -0.50), and the quadratic coefficient differs at a comparable level, so the loop sum differs from the converted one by several percent of P_{gg} at $k \simeq 0.1\,h\,{\rm Mpc}^{-1}$, the size of the whole correction being modeled. The linear column, read as an Eulerian bias, is low by 30–60% for the LRG and ELG chunks; a free amplitude absorbs that in a fit, while a fixed-fiducial use (a data vector, a covariance, a Fisher derivative) carries it.
- $b_3^{\rm L} = 0$ **is safe**, and is DESI's own baseline. The co-evolution value it discards is $|b_{3{\rm nl}}| \leq 0.17$, worth at most 0.6% of P_{gg} below $0.1\,h\,{\rm Mpc}^{-1}$, degenerate with α_0; Maus et al. [6] validate the choice directly. The drop is safe at both cuts provided α_0 stays in the model.

- $\alpha_0 k^2 P_{\rm lin}$ **against** $b_{k^2} k^2 P_{\rm mm}$: the same physical term in two dressings. They differ by $P_{\rm lin}$ versus $P_{\rm mm}$ (a 10–20% relabeling of the coefficient below $0.2\,h\,\mathrm{Mpc}^{-1}$) and by DESI's convention of factoring $(b_1 + f\mu^2)\, b_1\, \alpha_0$ out of the tree-level structure. CosmoLike's concrete form is $2\, b_1\, b_K\, k^2\, P_{\rm mm}$ with b_K the per-bin derivative-bias slot, so matching the draft's term needs $\alpha_0 = 2\, b_1\, b_K$ at the level of the k^2 coefficient plus the $P_{\rm lin}$-versus-$P_{\rm mm}$ relabeling. Any of the three is fine; the prior must be stated in the convention used, and the DESI width translates to a 50% correction to the linear spectrum at the analysis $k_{\rm max}$.

- **No redshift-space loops**: the draft keeps the loop sum in real space and attaches linear Kaiser projection plus the Fingers-of-God $n(z)$ convolution. The dropped μ-dependent loop and counterterm pieces are the thin-shell analogue of what DESI's α_2 and the one-loop velocity terms carry, and Section 5.2 shows thin shells do not suppress them. At $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ they are percent-level and partially degenerate with the sampled σ_v; the residual after marginalizing σ_v has not been measured, and that measurement (a Fisher bias of a data vector with redshift-space loops against the real-space-loop model) belongs to the implementation study. At $0.18\,h\,\mathrm{Mpc}^{-1}$ the drop is not defensible without that measurement, and the model needs an explicit radial counterterm.

- **No stochastic terms**: the Poisson part sits correctly in the covariance, but the non-Poisson residual is detected at tens of percent of Poisson for red samples [9], which is percent-level in P_{gg} for the LRG and ELG chunks (Figure 5). A forecast that fixes it to zero converts halo-exclusion physics into cosmological information. One SN_0 per chunk is required at either cut; the `sub_lowk` constants fold into the same parameter.

- **One constant bias set per DESI chunk**: the draft fixes (b_1, b_2, b_{s^2}) per chunk while the pipeline evolves $b_1 = B_g/D(z)$ across the chunk's thin bins. The two statements disagree with each other, and the constant option disagrees with co-evolution: if b_1 drifts by 7–20% across a chunk, b_{s^2} drifts with it (Figure 1, the b_{s^2} panel). Section 8 resolves this.

- **Halofit** $P_{\delta\delta}$: the hybrid is validated for projected statistics to beyond the shipped mask [9]. The requirements are consistency (the same $P_{\rm mm}$ in the data vector, covariance and Fisher derivatives) and awareness that halofit's accuracy, not the bias expansion, can set the error budget at the mask edge.

8 Bias evolution within a group

The project's correlated-prior note (`notes/correlated_bias_prior_v2`) models the linear bias of bin i in group g with that note's memory model (its section "The memory model"), a free tilt per group with a correlated prior on the residuals around the tilted line:

$$b_{1,i} = \frac{B_g\,(1+\delta_i)}{D(z_i)}, \qquad \delta_i = s_g\,(z_i - \bar{z}_g) + \varepsilon_i, \tag{16}$$

where z_i is the bin redshift, $\bar{z}_g$ the mean redshift of the group's bins, δ_i the fractional deviation from the group trend B_g/D, s_g the tilt of group g, free with a flat prior like B_g, and ε_i the residual. The residuals carry a Gaussian prior correlated across the bins of a group,

$$\langle \varepsilon_i\, \varepsilon_j \rangle = \sigma_g^2\, \exp\!\left(-\frac{|z_i - z_j|}{L}\right), \tag{17}$$

and zero across groups, with σ_g the per-bin prior width and L the correlation length in redshift; Section 9 fixes the baseline and pessimistic values of (σ_g, L).

8.1 Second-order trends from the per-bin linear bias

The higher-order sector should follow the same logic. Co-evolution makes the trend deterministic: given the per-bin linear biases $b_{1,i}$,

$$b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1) + \Delta_{s,g}, \qquad b_{2,i} = b_2^{\rm fit}(b_{1,i}) + \Delta_{2,g}, \tag{18}$$

with $b_{2,i}$ and $b_{s^2,i}$ the quadratic and tidal biases of bin i, $b_2^{\rm fit}$ the Lazeyras relation of Eq. (7) and $\Delta_{2,g}$, $\Delta_{s,g}$ one free offset per group.

The trends follow the full $b_{1,i}$, deviations included, not the smooth group shape $B_g/D(z_i)$. The reason is physical: a deviation δ_i is a real change of which halos the selection reaches in that bin (a flux limit, for example, keeps only the brighter galaxies, which sit in more massive halos), and the calibrated relations map halo populations at a given linear bias, so the second-order biases of the bin's population sit on the calibration curves at $b_{1,i}$, not at the group trend. The induced shift is small but determined: near $b_1 \simeq 2$ the Lazeyras relation has slope $db_2/db_1 \simeq 1.7$, so a residual at the per-bin width $\sigma_g = 3\%$ (the BGS baseline and the pessimistic width of every group) moves b_2 by about 0.10, a tenth of the offset prior width of 1.0 that Section 9 assigns, and the co-evolution slope $-4/7$ moves b_{s^2} by about 0.03 against the offset width of 0.5.

The offsets absorb the known imperfection of the calibrated relations (the 30% spread between co-evolution and the excursion-set γ_2 at LRG biases, and the scatter of the Lazeyras fit); the deterministic part carries the within-group evolution, which is not small: across LRG3, at $\delta_i = 0$, the co-evolution b_{s^2} moves from -0.88 to -1.09, a drift comparable to the offset prior width itself. Holding b_2 and b_{s^2} constant while $b_{1,i}$ evolves across the group is internally inconsistent; freeing them per bin throws away the physical correlation the correlated-prior note was written to keep. The next subsection grades the freedom between those two extremes.

8.2 A ladder for the higher-order freedom

Equation (18) already carries the per-bin drift of the higher-order sector: the calibration relations are evaluated at the full $b_{1,i}$ of Eq. (16), tilt and residuals included, so $b_{2,i}$ and $b_{s^2,i}$ inherit every move of the linear bias. What a free parameter adds on top is a drift of the calibration itself: b_2 at fixed b_1 changes with the satellite fraction and the color mix of the sample, and within one selection that change is slow. The freedom therefore sits per group, not per bin, and the choices form a ladder, from the most rigid to the most free:

- **Upper limit:** the calibration relations exactly, with no offsets. Zero parameters; every calibration error becomes a cosmology error.
- **Baseline:** the inherited trend plus one free offset per group, the $\Delta_{2,g}$ and $\Delta_{s,g}$ of Eq. (18): twelve parameters for the six chunks.
- **Pessimistic:** the inherited trend plus a free offset and a free tilt per group, $\Delta_g \to \Delta_g + t_g\,(z_i - \bar{z}_g)$ with t_g the offset tilt: 24 parameters.

	A baseline	A pessimistic	B baseline	B pessimistic
scale cut $k_{\max}$ [h/Mpc]	0.075	0.075	≈0.18	≈0.18
linear bias, tilt per chunk (new)	6	6	6	6
linear bias, per-bin residuals held by the prior	64 (smooth)	64 (rough)	64 (smooth)	64 (rough)
b_2, b_{s^2} offsets	12	24 (+tilt)	12	24 (+tilt)
α_0	6	12 (+tilt)	6	12 (+tilt)
α_2	—	—	6	12 (+tilt)
SN_0	6	6	6	6
SN_2	—	—	6	6
new parameters	94	112	106	130
bias-sector parameters without a tight prior, total including the 6 existing amplitudes	36	54	48	72

Table 1: The bias-sector parameter budget of the four configurations. The 64 residuals are the 76 per-bin deviations of Eq. (16) minus the free amplitude and tilt directions of the six chunks; "smooth" and "rough" name the baseline and pessimistic kernels of Eq. (17), whose (σ_g, L) values Section 9 fixes. The last row counts the 6 existing amplitudes B_g, so it is not a subset of the "new" row.

- **Lower limit:** b_2 and b_{s^2} free per bin, 152 parameters for the 76 bins. This option is a bound, not a candidate model. The thin-bin data cannot measure b_2 bin by bin, because one bin's loop terms are nearly degenerate with that bin's linear bias and counterterm, so the prior, not the data, would set the per-bin posteriors; and the freedom erases the scale information that justifies fitting to $k \simeq 0.18\,h\,\mathrm{Mpc}^{-1}$ in the first place.

The counterterms follow the same two middle options: one α_0 per group in the baseline (in model B, one α_2 as well), plus a tilt per group in the pessimistic case, because the counterterm scale follows the size of the host halos (their Lagrangian radius), a population property that drifts with the selection as the calibrations do. The stochastic amplitudes SN_0 and, in model B, SN_2 stay at one per group in both cases, with no tilt: the Poisson part $1/\bar{n}(z)$, with $\bar{n}$ the tracer number density, is known per bin and sits in the covariance, so only the non-Poisson fraction could drift, and the ladder tilts only what has a physical driver.

The correlated prior extends naturally to the offsets: chunks of the same tracer (LRG1–3, ELG1–2) share selection physics, so a Gaussian prior correlated across those chunks, with the same exponential kernel in redshift as Eq. (17), encodes "neighboring chunks deviate together" at second order.

Table 1 counts the parameters of the four configurations that Section 9 assembles: model A at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and model B at the shipped mask, each on the baseline and the pessimistic option. For reference, the DESI DR1 full-shape analysis uses seven nuisance parameters per tracer chunk, 42 in total at $k_{\max} = 0.2\,h\,\mathrm{Mpc}^{-1}$ [7, 8], so model B baseline is that budget plus one b_1 tilt per chunk.

9 Recommendation

Both models below are one-loop models. Each carries the complete real-space one-loop Eulerian bias set of Eq. (2) (b_1, b_2, b_{s^2}, $b_{3\mathrm{nl}}$), the k^2 counterterm α_0 and the stochastic amplitude SN_0; they differ in what is free. Model A ties the higher-order biases to the calibrations through the

per-bin linear bias, with group offsets under modest priors; model B frees them and adds the redshift-space terms.

At $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ (the cut matched to the thin-bin radial resolution), the minimal defensible model per DESI chunk, called model A below, is

- $b_{1,i} = B_g(1+\delta_i)/D(z_i)$ per thin bin, the memory model of the correlated-prior note (Eqs. (16) and (17)): the amplitude B_g and the tilt s_g free with flat priors, one each per group, twelve parameters in total, and the residuals ε_i under the exponential-kernel prior. The baseline sets $\sigma_g = 3\%$ for BGS, sized to its flux-limit curvature, $\sigma_g = 1\%$ for the other groups, and $L = 0.2$. The pessimistic choice sets $\sigma_g = 3\%$ in every group and $L = 0.05$, the redshift scale of local selection features such as the 4000 Å break crossing a filter edge;
- $b_{2,i}$ from the Lazeyras relation on $b_{1,i}$ (Eq. (18)) plus a free offset $\Delta_{2,g}$, Gaussian prior of width 1.0 centered on zero;
- $b_{s^2,i}$ from co-evolution on $b_{1,i}$ plus a free offset $\Delta_{s,g}$, Gaussian prior of width 0.5 centered on zero (the width covers the co-evolution-versus-excursion-set spread at LRG biases);
- $b_{\rm 3nl}$ fixed to co-evolution (or zero; the difference is inside the α_0 freedom);
- one counterterm α_0 per chunk, Gaussian prior centered on zero with the DESI rule rescaled to this cut: a 50% correction at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ means a width of $0.5/k_{\rm max}^2 \simeq 89\,h^{-2}{\rm Mpc}^2$;
- one stochastic amplitude SN_0 per chunk, Gaussian prior of width twice the Poisson power, following DESI;
- the sampled σ_v groups kept as the radial damping freedom, with the implementation study measuring whether they cover the dropped μ-dependent terms at this cut.

The linear sector carries one amplitude per thin bin (76 in the shipped layout of equal comoving bin width $\Delta d_{\rm com} = 62\,{\rm Mpc}$), constrained by the kernel prior in every direction except the flat amplitude B_g and tilt s_g of each group: twelve flat directions, with the remaining 64 residual combinations held by the prior. The 24 higher-order parameters (four per chunk: $\Delta_{2,g}$, $\Delta_{s,g}$, α_0, SN_0) come on top, against DESI DR1's seven per tracer at $k_{\rm max} = 0.2\,h\,{\rm Mpc}^{-1}$; the pessimistic option of Section 8.2 tilts the two offsets and α_0 and reaches 42, seven per chunk (Table 1). The reasoning: at this cut every correction is 1–3% with mild scale dependence, so calibration-centered Gaussian freedom is enough, and the offsets keep the model honest about the calibrations without opening flat directions.

The growth-tied linear bias $b_1(z) = B_g/D(z)$ with no deviations, the shape the shipped parameter files tie to, is not a defensible alternative. The correlated-prior note shows that even a conserved population departs from that shape, by about 1% per $\Delta z = 0.05$ through the bias continuity equation, and that selected samples drift more: the BGS flux limit moves the bias by about 22% across its group (its sections "The bias continuity equation, step by step" and "The BGS flux-limit bias model, step by step"). The linear-bias fits of `notes/linear_bias_kmax` measure what the tying buys in the dark-energy figure of merit ${\rm FoM}(w_0, w_a)$, the inverse area of the error ellipse of the equation-of-state parameters w_0 and w_a, on DESI C_ℓ^{gg} alone at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ and the $\Delta d_{\rm com} = 62\,{\rm Mpc}$ layout: the growth-tied prior gives 32.9, the baseline prior 18.2, the pessimistic prior 11.4, and the free prior, one free amplitude per bin, 10.7. The factor of 1.8 between the growth-tied and baseline-prior numbers rests on a rigidity the physics

does not support: in the accuracy test of the linear-bias priors (defined in the correlated-prior note's section "The accuracy test: the figure of bias"), a conserved galaxy population fitted with the growth-tied prior displaces (w_0, w_a) by 11–21σ, while the baseline and pessimistic priors keep the displacement below $\sigma/3$.

At $k_{\rm max} \simeq 0.15$–$0.2\,h\,{\rm Mpc}^{-1}$ (the shipped mask), the model, called model B below, keeps the per-bin linear bias $b_{1,i}$ with the memory model and widens the higher-order sector: the b_2 and b_{s^2} offsets with the calibration relations demoted to prior centers (widths of order 2 and 1), α_0 at the DESI DR1 width $12.5\,h^{-2}{\rm Mpc}^2$, an explicit radial counterterm (the $\alpha_2\mu^2k^2P_{\rm lin}$ term projected through the j_ℓ'' transfer, or a validated σ_v-based equivalent) at the same DESI DR1 width, SN_0 at twice the Poisson power, SN_2 at the DESI DR1 width of five units of $f_{\rm sat}\sigma_v^2/\bar{n}$ (Section 3), and the redshift-space one-loop question settled by measurement rather than assumption. Counting the linear amplitude and tilt, the baseline option is eight parameters per chunk: the DESI DR1 budget of seven plus one b_1 tilt; the pessimistic option also tilts the offsets and both counterterms (Table 1). The honest statement is that the real-space-loop shortcut is unvalidated at this cut: until a Fisher-bias test shows otherwise, headline numbers from the shipped mask carry a modeling systematic of the size of the α_2-equivalent terms. Sampling in a physical basis ($b_1\sigma_8$-like combinations) is worth adopting at either cut, for DESI's reason: it tracks what the data constrain and reduces projection effects.

In both configurations, the fiducial higher-order values are the co-evolution and Lazeyras values of Figure 1, not imports of DESI chain values, for the reason Section 4 ends with; and the covariance and Fisher machinery must adopt the same model as the data vector, including the stochastic sector.

10 Implementation in CoCoA

This section reports the implementation study that ran alongside this note: an inventory of what CoCoA/CosmoLike computes today for one-loop galaxy bias, the relation of the draft's CCL data vector to it, and the gaps against models A and B of Section 9. The study itself changed no code: the inventory and the gap table describe the code as the study found it, and the status list after the table records the model A work done since. File references are to `external_modules/code/cosmolike_core/cosmolike` (the `lowrank_fftlog` branch) and to `projects/lssty1xdesi`.

10.1 What CosmoLike computes

The one-loop machinery of Section 2 largely exists in the C code; what is missing sits at the edges (the cross pairs, the stochastic sector) and in what the project likelihood passes.

- **Loop templates.** `get_FPT_bias` in `pt_cfastpt.c` computes the five quadratic templates $P_{\delta_1\delta_2}$, $P_{\delta_2\delta_2}$, $P_{\delta_1 s_2}$, $P_{\delta_2 s_2}$, $P_{s_2 s_2}$ from thirteen C-FAST-PT J terms at $z = 0$, rescaled by D^4 inside the projection integrands, exactly the separable structure of Section 5.1; the tables are cosmology-dependent and rebuilt when the cosmology changes. A second producer, `set_bias_PS` in `generic_interface.cpp`, can install a Python FAST-PT table instead. The $b_{3\rm nl}$ template $P_{\delta_1\psi}$ is the exception: `tab_d1d3` in `pt_cfastpt.h` is a fixed, cosmology-independent table, acceptable only while $b_{3\rm nl} = 0$.

- **Low-k constants.** The galaxy-clustering Limber integrand in `cosmo2D.c` subtracts the white-noise constants of Section 2.1 ($P_{\delta_2\delta_2} - 2\sigma_4$ and partners, with σ_4 read as half the low-k limit of $P_{\delta_2\delta_2}$): the `sub_lowk`-on convention, against CCL's default.

- **Bias slots and spectra.** `bias.c` carries per-bin parameters `gb1`, `gb2`, `gbs2`, `gb3`, `gbmag`, `gbK`. `C_gg_tomo_limber_work` assembles the auto integrand $b_1^2 P_{\rm mm} + 2\, b_1 b_K k^2 P_{\rm mm}$ plus D^4 times the six loop terms of Eq. (2); `C_gs_tomo_limber_work` and `C_gk_tomo_limber_work` carry the matching galaxy–lensing cross terms ($\frac{1}{2} b_2 P_{\delta_1\delta_2} + \frac{1}{2} b_{s^2} P_{\delta_1 s_2} + \frac{1}{2} b_{\rm 3nl} P_{\delta_1\psi} + b_K k^2 P_{\rm mm}$).

- **Non-Limber engines.** `C_gg_lowrank_decomp_work`, the low-rank engine, builds its per-bin SVD of $\sqrt{P_g}$ from the complete one-loop Limber integrand, so the auto spectra are already exact in the one-loop P_g; it clamps $P_g \geq 10^{-15}\, b_1^2 P_\delta$ where the loop sum turns negative, the guard Section 5.1 discusses, since replaced by the exact treatment of Section 5.4. The classic FKEM core (`C_cl_tomo`) projects only $b_1 P_{\rm lin}$ exactly and moves everything else through its Limber difference, so its one-loop content is Limber-only: the failure mode with scale-dependent bias at thin bins that the draft paper describes. The adjacent-bin cross pairs (`C_gg_cross_tomo_limber_work`) stop with `log_fatal` when second-order bias is active: not implemented.

- **Redshift-space distortions and stochastic terms.** Linear Kaiser projection (the j_ℓ'' transfer) plus the σ_v Fingers-of-God smearing of each $n(z)$; no one-loop redshift-space terms, no μ-dependent counterterms, and no SN_0 or SN_2 slots anywhere.

- **Interface and likelihood.** `set_nuisance_bias_fastpt` in `generic_interface.cpp` (arguments `B1`, `B2`, `B_MAG`, `B3nl`, `BK`), bound as `set_nuisance_bias` in the project's `interface/interface.cpp`, takes no tidal argument: `set_nuisance_nonlinear_bias` sets $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ per bin whenever $b_2 \neq 0$, the co-evolution relation of Eq. (6) in the CCL normalization. The project likelihood (`likelihood/desi_cl.py`) passes linear bias only (`B2` = `B3nl` = `BK` = `B_MAG` = 0), so none of the machinery above is exercised by lssty1xdesi today.

- **Fisher and covariance.** The Fisher module, `cosmolike_lssty1xdesi_fisher.py` under `interface/`, varies every declared `DESI_` parameter, so new nuisances enter it automatically once the likelihood reads them; C_ℓ is linear in b_2, b_{s^2}, b_K and SN_0 at fixed b_1 apart from the quadratic $\frac{1}{4} b_2^2$ and $\frac{1}{4} b_{s^2}^2$ terms, so the new derivatives are cheap. With b_2 and b_{s^2} following $b_{1,i}$ through Eq. (18), the derivative with respect to a deviation δ_i also picks up the chain through b_2 and b_{s^2}: the loop terms move with $db_2/db_1 \simeq 1.7$ and $db_{s^2}/db_1 = -4/7$ times the change of $b_{1,i}$. The covariance (`covariances/spectra_cov.c`) carries linear bias only, on top of the open mismatch that it takes the raw chunk amplitudes instead of $B_g/D(z)$.

10.2 The draft's data vector against CosmoLike

The two calculations share the six templates in the same normalization (CCL, FAST-PT and CosmoLike agree term by term). They differ in three places. First, the low-k constants: CosmoLike subtracts them, and a CCL vector at the CCL default does not, so the two vectors differ by the white-noise term quantified in Section 2.1 until the draft states its `sub_lowk` setting. Second, the k^2 term: the draft adds $\alpha_0 k^2 P_{\rm lin}$ while CosmoLike has $2\, b_1 b_K k^2 P_{\rm mm}$; matching them needs

$\alpha_0 = 2\,b_1 b_K$ at the level of the k^2 coefficient plus the $P_{\rm lin}$-versus-$P_{\rm mm}$ relabeling (Section 7). Third, the tidal bias: the interface fixes b_{s^2} to co-evolution, so CosmoLike cannot take the draft table's tidal values; given Section 4, the co-evolution default is closer to the physical value than those entries. The implementation study's independent check of that verdict is folded into Section 4.

10.3 Gaps against model A

Model A (Section 9: the per-bin linear bias $b_{1,i}$ with the free tilt and the kernel prior; per chunk, the b_2 and b_{s^2} trends on $b_{1,i}$ with group offsets, $b_{3\rm nl}$ from co-evolution or zero, one α_0, one SN_0, the σ_v groups kept) decomposes into the following work items.

need	status	change
per-bin linear bias with the memory model	the likelihood reads one `DESI_B1_`n per thin bin; the shipped yamls tie them per group; the Fisher notebook already applies the kernel and memory prior precisions	a yaml grouping that samples every bin and one tilt per group, plus a cobaya external prior carrying the kernel Gaussian on the residuals (small, Python and yaml)
b_2 trend + offset per group	per-bin `gb2` exists in C	likelihood only: compute per-bin values from $b_{1,i}$ and a new `DESI_B2OFF` parameter per group in `desi_cl.py` (small, Python)
b_{s^2} trend + offset	C slot exists; the interface forces co-evolution	add a tidal argument to `set_nuisance_bias`, keeping the co-evolution default when absent; likelihood computes the per-bin values (small, C++ interface and Python)
$b_{3\mathrm{nl}}$	fixed template table	keep $b_{3\mathrm{nl}} = 0$ (the DESI baseline); make `tab_d1d3` cosmology-dependent before ever freeing it
α_0 counterterm	$2\,b_1 b_K k^2 P_{\mathrm{mm}}$ exists	reuse b_K with the stated mapping, or add a $k^2 P_{\mathrm{lin}}$ variant for an exact match to the draft and DESI (small)
SN_0 per chunk	missing	new nuisance slot; the projection is the exact Eq. (14), autos plus the overlap of Fingers-of-God-convolved adjacent kernels (small, C)
non-Limber autos	low-rank engine already exact in the one-loop P_g	none
non-Limber adjacent cross pairs	`log_fatal` with second-order bias	the main C task: each template separates as $D^2(z_1)D^2(z_2)$ (Section 5.1), so one extra FFTLog projection per template, or an extension of the low-rank cross factors to the one-loop P_g, with the Limber subtraction rule unchanged
classic FKEM	one-loop content Limber-only	use the low-rank engine on thin bins; document that classic FKEM is not valid with one-loop bias there
covariance	linear bias only	add the one-loop P_{gg} together with the $B_g/D(z)$ bias fix to `spectra_cov.c` (medium)
Fisher	prefix-based	none beyond the likelihood parameters

Status of the gaps. The table records the starting point; the items below record the status of the code-side gaps after the model A work.

- **Model A: implemented end to end.** The `desi_cl` likelihood runs model A with the option `higher_order_bias: 1`, and the example yamls `EXAMPLE_EVALUATE_MODEL_A1–A4.yaml` set it up on the four bin layouts. The Fisher notebook `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb` builds its Fisher matrices, and `notes/correlated_bias_prior_v2` reports the bias ladder and its accuracy test under model A.

- **Adjacent-bin cross pairs: implemented** in cosmolike_core, where the pair spectra had stopped with `log_fatal` under second-order bias. The full output carries the bilinear one-loop pair spectrum: the auto's one-loop sum with each product of two biases shared between the two bins. The density-only output, which the low-rank splice subtracts, is the geometric mean $W_i W_j \sqrt{P_{g,i} P_{g,j}}$, the high-$\ell$ limit of the low-rank pair term. Checks: linear-bias vectors are bitwise unchanged with the low-rank, FKEM and Limber engines; a pair of two bins with identical $n(z)$ equals the polarization of the auto spectra (the bilinear form recovered from the quadratic one) to 10^{-14} of the one-loop increment; 1 and 8 threads give identical vectors.

- **Non-Limber autos: exact** (the cosmolike_core change described in Section 5.4). The low-rank engine projects the white-noise constant without a floor (Section 5.4).

- **Tidal bias: implemented.** A new setter, `set_nuisance_bias_fastpt_tidal`, takes b_{s^2} per bin; the co-evolution setter is unchanged.

- SN_0**: implemented** in `desi_cl.py` as the exact projection of Eq. (14) with the Fingers-of-God-convolved kernels, in units of each bin's Poisson power. Cross pairs receive it only within one tracer sample: an LRG and an ELG bin at the same redshift are different galaxies and share no stochastic term.

- α_0**: passed through the derivative-bias slot** as $b_K = \alpha_0/[2b_1(c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}\mathrm{Mpc}$, because CosmoLike's k is in units of H_0/c.

- α_0 **response: the low-rank engine is required.** FKEM moves every nonlinear term, the counterterm included, through its Limber difference, while the low-rank engine projects the counterterm without the Limber approximation. On the $\Delta z = 0.05$ layout at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ (at $z = 0$), the low-rank α_0 response is 7.3 times the Limber one at $\ell = 20$ and 1.26 times at $\ell = 100$ (median over the kept autos), from the k^2-weighted radial modes of thin, sharp-edged bins. It is converged, and the model A forecast uses this engine.

- α_0 **beyond the cut: open.** On the same layout the ultraviolet fraction of the α_0 template on the kept bands is a median 0.076 (maximum 0.14), above $\alpha = 0.05$ on every kept band, against a median 0.017 for the fiducial model. The cut was built on the linear model; whether to rebuild it on the one-loop model, templates included, is an open decision.

- **Negative** α_0 **in chains: open.** At $\alpha_0 = -88.9\,h^{-2}\mathrm{Mpc}^2$, one prior width below zero, $P_{gg} - P_\infty$ turns negative above $k \approx b_1/\sqrt{|\alpha_0|} \approx 0.14\,h\,\mathrm{Mpc}^{-1}$ and the low-rank engine floors 47% of its grid, so its vector curves in α_0 while FKEM's stays linear. A Fisher forecast at $\alpha_0 = 0$ is unaffected; chains far from $\alpha_0 = 0$ need the loop taper or the positivity prior of Section 5.4. The appendix "The counterterm: engines, scale cut and sign" of `notes/correlated_bias_prior_v2` gives the numbers of these three items.

- $b_{3\mathrm{nl}}$**: zero.** Model A sets $b_{3\mathrm{nl}} = 0$ and requires the likelihood option `bias_model` to be $[0, 0, 0, 0, 0, 0]$ (a code of 1 sets $b_{3\mathrm{nl}} = b_1 - 1$). CosmoLike's $b_{3\mathrm{nl}}$ template `tab_d1d3` is a fixed, cosmology-independent table that disagrees with Python FAST-PT: $\Delta\chi^2 \approx 3.4$ on the $\Delta d_{\mathrm{com}} = 62\,\mathrm{Mpc}$ vector with $b_{3\mathrm{nl}} = b_1 - 1$.

- **FAST-PT reference.** With $b_{3\mathrm{nl}} = 0$, C-FAST-PT and Python FAST-PT agree to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ on the $\Delta d_{\mathrm{com}} = 62\,\mathrm{Mpc}$ thin-bin vector, pairs included, so the C-FAST-PT sampling needs no retuning. The comparison ran through the `fastpt` theory block (`IA_code` 1) under the TATT intrinsic-alignment model (tidal alignment and tidal torquing), which for DESI C_ℓ^{gg} is equivalent to NLA (nonlinear alignment). Convergence: doubling the C-FAST-PT internal grid gives $\Delta\chi^2 \sim 10^{-8}$; `accuracyboost` 3–4 gives $\Delta\chi^2 \leq 4.5 \times 10^{-3}$ on the one-loop part; the low-rank rank is converged at 5 (rank 5 to 12 moves $\Delta\chi^2$ by 10^{-10}). Open bug: `internal_accuracyboost` $= 4$ breaks the C-FAST-PT bias tables ($\Delta\chi^2 \sim 10^{12}$).

- **Covariance: open.** It still carries linear bias.

10.4 Model B and the order of work

Model B widens the b_2 and b_{s^2} offset priors to the demoted calibrations (yaml work once model A is in) and adds the radial counterterm α_2, projected with the j_ℓ'' transfer like the Kaiser term, at the DESI DR1 width $12.5\,h^{-2}\mathrm{Mpc}^2$, the stochastic term SN_2 at the DESI DR1 width of five units of $f_{\mathrm{sat}}\sigma_v^2/\bar{n}$, and the redshift-space one-loop terms. CosmoLike has no redshift-space loop templates and FAST-PT's redshift-space terms are not in the C port, so the options are: absorb them in the sampled σ_v plus an explicit α_2-like counterterm and measure the Fisher bias against a full redshift-space model, or port the FAST-PT redshift-space templates (or an external $P(k, \mu)$ emulator). Model B is a research task, not an extension of model A.

The recommended order:

1. yamls: a grouping that samples every `DESI_B1_`n and one tilt per group, plus the cobaya external prior carrying the kernel Gaussian on the residuals for chains;

2. interface: free the tidal argument (co-evolution default), add the SN_0 slot, state the b_K-to-α_0 mapping;

3. likelihood: the b_2 and b_{s^2} trends on $b_{1,i}$ with group offsets, α_0, SN_0, with the priors of Section 9;

4. C: one-loop for the adjacent cross pairs through the separable $D^2(z_1)D^2(z_2)$ templates;

5. validation: CosmoLike's Limber one-loop against CCL's `EulerianPTCalculator` with the same templates and the same `sub_lowk` choice (the `ccl` conda environment carries pyccl and FAST-PT; `cocoapy311` carries neither), then the low-rank non-Limber against a brute-force Bessel integral for one thin pair;

6. covariance: the one-loop P_{gg} and the $B_g/D(z)$ bias fix together;

7. model B only after a measured Fisher bias of the redshift-space loops.

A Bases and conversions, with every factor

A.1 The operators

The tidal field of a density field δ is $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\delta$, with $s^2 = s_{ij}s_{ij}$. Writing the normalized potential $\phi = \partial^{-2}\delta$, expanding the square gives $s^2 = (\partial_i\partial_j\phi)^2 - \frac{2}{3}\,\delta\,\partial^2\phi + \frac{1}{9}\,\delta_{ij}\delta_{ij}\,\delta^2 = (\partial_i\partial_j\phi)^2 - \frac{1}{3}\,\delta^2$, using $\partial^2\phi = \delta$ and $\delta_{ij}\delta_{ij} = 3$. The Galileon operator is $\mathcal{G}_2 = (\partial_i\partial_j\phi)^2 - \delta^2$, so

$$\mathcal{G}_2 = s^2 - \frac{2}{3}\,\delta^2. \tag{19}$$

A term $b_{\mathcal{G}_2}\mathcal{G}_2$ in δ_g therefore equals $b_{\mathcal{G}_2}s^2 - \frac{2}{3}b_{\mathcal{G}_2}\delta^2$: matching to Eq. (1), $b_{s^2} = 2\,b_{\mathcal{G}_2}$, and the quadratic coefficients of the two bases relate as $b_2 = b_2' - \frac{4}{3}\,b_{\mathcal{G}_2}$, where b_2' is the coefficient of $\delta^2/2$ in the $(b_2', b_{\mathcal{G}_2})$ basis. The third-order operator of Eq. (1) is $\psi = \eta - \frac{2}{7}s^2 + \frac{4}{21}\delta^2$ with $\eta = \theta - \delta$ the velocity-divergence difference [1, 5]; at one loop only one combination of the cubic operators survives renormalization, with the filter $\sigma_3^2(k)$ of [2] as its power-spectrum template.

A.2 Normalization bookkeeping

Three conventions circulate for the quadratic and tidal terms, and every difference discussed in Section 4 is a crossing between two of them:

convention	quadratic term	tidal term
CCL / FAST-PT / [2]	$(b_2/2)\,\delta^2$	$(b_{s^2}/2)\,s^2$
velocileptors Eulerian [5]	$(c_2/2)\,\delta^2$	$c_s\,s^2$
velocileptors Lagrangian [5, 6]	$(b_2^{\rm L}/2)\,\delta_0^2$	$b_s^{\rm L}\,s_0^2$

So $b_2 = c_2$ but $b_{s^2} = 2c_s$: the quadratic parameters agree across the Eulerian conventions while the tidal ones differ by the factor of two that Eq. (5) carries.

A.3 The rotation, from the co-evolution solution

The 8/21 and $-2/7$ of Eq. (4) follow from solving the coupled continuity and Euler equations for a tracer that starts with local Lagrangian bias and is then advected (the derivation followed here is [2]; the same coefficients appear as the bias-parameter rotation in [5]). At linear order the Lagrangian bias dilutes as $b_1^{\rm L}(z) \propto 1/D(z)$, equivalently $b_1(z) - 1 \propto 1/D(z)$: the co-evolution statement and the project's constant-clustering-amplitude shape $B_g/D(z)$ coincide for the linear bias up to the additive 1. At second order the evolved overdensity picks up, next to the gravitational F_2 kernel, the terms

$$\delta_g^{(2)} \supset \left[\frac{b_2^{\rm L}}{2} + \frac{4}{21}b_1^{\rm L}\right]\delta^2 \;-\; \frac{2}{7}\,b_1^{\rm L}\,s^2, \tag{20}$$

which, matched to $(c_2/2)\delta^2 + c_s s^2$, give $c_2 = b_2^{\rm L} + \frac{8}{21}b_1^{\rm L}$ and $c_s = b_s^{\rm L} - \frac{2}{7}b_1^{\rm L}$ once an initial $b_s^{\rm L}s_0^2$ is carried along. At third order the same procedure yields the dynamically generated piece $b_{3\rm nl} = \frac{32}{315}\,b_1^{\rm L}$ of Eq. (6) [2]. In the CCL normalization the tidal entries double, as Eq. (5) records.

A.4 Worked example: LRG1

DESI samples $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$, $b_s^{\rm L}\sigma_8^2$ at $z_{\rm eff}=0.510$. Suppose the chain value of the first is 1.448 and $\sigma_8(0.510)=0.649$ at the fiducial cosmology. Then $b_1=1.448/0.649=2.23$, and the co-evolution fiducials for the forecast are $b_{s^2}=-\frac{4}{7}(1.23)=-0.70$ and $b_{3\rm nl}=\frac{32}{315}(1.23)=+0.125$, with $b_2(b_1{=}2.23)=+0.34$ from Eq. (7). Using 1.448 itself as an Eulerian bias, as the draft table does, is a 31% error on b_1 and a factor b_1^2 error of 69% on the leading term of P_{gg}, before the tidal-sign error enters.

B The loop templates and their low-k constants

With F_2 and $S_2=\mu_{12}^2-\frac{1}{3}$ the standard second-order density and tidal kernels (μ_{12} the cosine between the two internal wavevectors), the quadratic templates of Eq. (2) are

$$
\begin{aligned}
P_{\delta_1\delta_2}(k) &= 2\int_{\mathbf{q}} F_2(\mathbf{q},\mathbf{k}-\mathbf{q})\,P_{\rm lin}(q)\,P_{\rm lin}(|\mathbf{k}-\mathbf{q}|), & P_{\delta_2\delta_2}(k) &= 2\int_{\mathbf{q}} P_{\rm lin}\,P_{\rm lin},\\
P_{\delta_1 s_2}(k) &= 2\int_{\mathbf{q}} F_2\,S_2\,P_{\rm lin}\,P_{\rm lin}, & P_{\delta_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2\,P_{\rm lin}\,P_{\rm lin},\\
P_{s_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2^2\,P_{\rm lin}\,P_{\rm lin}, &&
\end{aligned}
\tag{21}
$$

(arguments as in the first line throughout), evaluated by FAST-PT as FFTLog convolutions. The $k\to 0$ limits are $P_{\delta_2\delta_2}\to 2\sigma_4$, $P_{\delta_2 s_2}\to\frac{4}{3}\sigma_4$, $P_{s_2 s_2}\to\frac{8}{9}\sigma_4$ with $\sigma_4=\int_{\mathbf{q}}P_{\rm lin}^2(q)$, white-noise constants that renormalize into the stochastic sector; CCL's `sub_lowk` subtracts exactly these three constants in the combinations shown in Eq. (2). The third-order template is $\sigma_3^2(k)P_{\rm lin}(k)$ with the filter σ_3^2 defined through the I_R kernel of [2]; its co-evolution-weighted contribution is the $b_1 b_{3\rm nl}$ curve of Figure 4. The unequal-time factors quoted in Section 5.1 follow from the perturbative order of each side of the correlator: 22-type products of two second-order fields carry $D^2(z_1)D^2(z_2)$, 13-type products of a third-order and a linear field carry the symmetrized D^3D.

C Numerical setup for the figures

Every figure of this note lives in its own folder under `plots/`, named after the figure, with its data (`data.json`) and the script that draws it from those data alone; `plots/make_figure_data.py` writes every `data.json`.

Figures 1, 2, 4 and 5. Cosmology: the project fiducial, $\Omega_m=0.3$, $\Omega_b=0.04$, $h=0.6732$, $10^9A_s=2.1$, $n_s=0.966$, massless neutrinos, giving $\sigma_8(0)=0.843$. Linear and halofit spectra from CAMB 1.6.5, halofit in the Takahashi version the likelihood uses; loop templates from FAST-PT 4.0 on a 1200-point logarithmic grid with low- and high-k extrapolation and a FFTLog window of 0.75, mirroring CCL's `EulerianPTCalculator` defaults; growth factors from the evolution of the $k=0.05\,{\rm Mpc}^{-1}$ mode. The packages come from the group's `ccl` conda environment (pyccl 3.3.3 is present there for cross-checks; the `cocoapy311` environment has neither fastpt nor pyccl). Thin-bin densities for Figure 5: the project files `data/lssty1xdesi_desi_dr3_ngal.txt` (surface density per bin) and `..._bins.txt` (bin mean redshifts), converted to three-dimensional densities with $\bar n=n_{2D}/(\chi^2\,\Delta\chi)$ at $\Delta\chi=62.28\,{\rm Mpc}$. The $\sigma_8(z_{\rm eff})$ values of Figure 2 use the

same cosmology; the DESI fiducial differs from it by a few percent in σ_8, which moves the implied biases coherently by the same few percent and does not affect the identification.

Figure 3 and the σ_4 of Eq. (12) use the likelihood's evaluation cosmology instead: the same parameters with $n_s = 0.96605$ and one massive neutrino of 0.06 eV inside Ω_m, the linear spectrum on a 2000-point logarithmic grid up to $40\,h\,\mathrm{Mpc}^{-1}$, and the growth factor from the ratio of the linear spectra at $k = 10^{-4}\,h\,\mathrm{Mpc}^{-1}$. The massless set-up above gives a σ_4 larger by 7% and values of $k_{\rm neg}$ within 2% of these.

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